

Unit 1 — Banking (Questions 1–12)

1. Explain the main functions of RBI.

Short note: RBI's main functions are monetary policy (control money supply & inflation), banker to government, banker's bank (lender of last resort, liquidity support), currency issue & management, regulation & supervision of banks, foreign exchange management (preserve external value), and development functions (promote financial inclusion, credit flow).

2. Draw the structure of Indian banking system.

Short note: Top — Reserve Bank of India (central bank). Next level — Public sector banks, Private sector banks, Foreign banks, Regional Rural Banks (RRBs), Cooperative banks, Small Finance and Payments banks. Parallel — Development banks (NABARD, SIDBI), NBFCs. (Use a simple pyramid diagram for exam answers.)

3. Differentiate between Scheduled and Non-Scheduled bank.

Short note: Scheduled banks appear in RBI's Second Schedule & meet minimum paid-up capital/reserves criteria; eligible for RBI refinance and access to clearing/other facilities. Non-scheduled banks do not meet criteria and don't have direct access to RBI's refinance.

4. What is commercial bank? Describe the different types of commercial banks.

Short note: Commercial banks accept deposits and provide loans to earn profit. Types: Public sector banks, Private sector banks, Foreign banks, Regional Rural Banks, Cooperative banks, Small Finance Banks, Payments banks, and Specialized banks.

5. Explain Co-operative banks, Development banks and NBFCs.

Short note: Co-op banks are member-owned, serve local communities. Development banks (e.g., NABARD, SIDBI) provide medium/long-term finance for development projects. NBFCs (non-banking financial companies) provide loans and credit but don't take demand deposits and are regulated differently (RBI oversight).

6. Discuss the primary and secondary functions of bank.

Short note: Primary functions — Accepting deposits (savings, current, fixed), advancing loans and discounts. Secondary functions — Agency

services (payments, collection), general utility services (locker, forex, issuing guarantees), investment services, credit creation.

7. Explain various types of Bank accounts.

Short note: Savings account (for individuals, interest-bearing), Current account (for businesses, no/low interest, overdraft), Fixed deposit (term deposits), Recurring deposit (monthly savings), NRI accounts (NRE, NRO, FCNR), Demat/trust accounts.

8. What is cheque? Explain the important elements of cheque and various types of cheque.

Short note: A cheque is a written order directing a bank to pay a specified sum from drawer's account to payee. Key elements: drawer, drawee bank, payee, date, amount (in words & figures), signature. Types: Bearer cheque, Order cheque, Crossed cheque, Open cheque, Stale cheque, Post-dated cheque, Ante-dated cheque.

9. Discuss important aspects related to Debit card, Forex card and Credit card.

Short note: Debit card draws funds directly from your account; PIN-secured; ATM withdrawals & PoS payments. Credit card provides short-term credit, interest if not paid by due date, rewards & EMIs.

Forex/prepaid travel card is preloaded in foreign currency — safer/cheaper FX rates, used abroad/online.

10. Explain all aspects related to personal loans and home loans.

Short note: Personal loans — unsecured, higher interest, short-tenure, used for personal needs. Home loans — secured by property, lower interest, long tenure, tax benefits on interest/principal (subject to rules). Both need eligibility checks (income, credit score), documentation, processing fee.

11. What is NPA? Explain classification of assets under NPA?

Short note: NPA = Non-Performing Asset — loan in which interest or principal is overdue >90 days (standard RBI norm, though some loan types have different norms). Classifications: Standard asset, Sub-standard (NPA ≤12 months), Doubtful (NPA >12 months with deteriorated recovery), Loss asset (virtually unrecoverable).

12. Explain all major factors affecting CIBIL Score and suggest how to improve CIBIL Score.

Short note: Factors: payment history (most important), credit utilization, length of credit history, types of credit, recent credit inquiries. Improve: pay EMIs on time, reduce credit utilization (<30%), avoid multiple loan applications, keep old credit active, clear outstanding defaults.

Unit 2 — Insurance (Questions 13–20)

13. Explain the role of insurance in detail.

Short note: Insurance transfers risk from individual/entity to insurer in exchange for premium. Roles: risk pooling, indemnity, financial protection, promoting savings (life insurance), enabling credit (collateral), encouraging investment by reducing uncertainty, and social security.

14. Discuss the Duties, Powers and Functions of IRDAI.

Short note: IRDAI (Insurance Regulatory & Development Authority of India) regulates insurance industry — grant licenses, protect policyholders, regulate rates/product filings, ensure solvency & capital requirements, promote competition, monitor distribution channels, enforce compliance.

15. Differentiate between Life Insurance and General Insurance.

Short note: Life insurance provides long-term protection for life — pays on death/longevity (also savings). General insurance (non-life) covers property, health, motor — short-term (usually annual) and indemnity-based.

16. Describe various types of Life Insurance.

Short note: Term insurance (pure risk cover), Endowment policies (risk + savings), Money-back, Whole life, ULIP (unit-linked insurance plan — investment + insurance), Pension/annuity products, Child plans.

17. Discuss various aspects related to life insurance such as policy premium, Policy Claim, Policy Lapsation, Revival, Nomination and Surrender of a Policy.

Short note: Premium = amount paid for cover. Claim = beneficiary files proof at death/claim event. Lapse = policy terminated for non-payment of premium (after grace period). Revival = reinstating lapsed policy with

arrears + conditions. Nomination = beneficiary designation. Surrender = voluntary termination for surrender value (subject to policy terms).

18. Explain Motor Insurance, and Travel Insurance in detail.

Short note: Motor insurance — third-party liability (mandatory) + own-damage cover (comprehensive). Covers accidents, theft, third-party claims. Travel insurance — covers trip cancellation, medical emergencies abroad, lost baggage, delays; limited by policy terms.

19. Explain Home Insurance, Personal Accident Insurance Policy, and Pradhan Mantri Suraksha Bima Yojana (PMSBY) in detail.

Short note: Home insurance covers house structure & contents against fire, theft, natural calamities. Personal accident covers accidental death/disablement. PMSBY — government-supported low-cost accident insurance (nominal premium; covers death/permanent disability) available to eligible bank account holders.

20. What is reinsurance? Explain importance, types and benefits of reinsurance.

Short note: Reinsurance = insurer transfers risk to another insurer (reinsurer). Importance: risk diversification, protect insurer solvency, capacity to underwrite large risks, stabilize loss experience. Types: treaty & facultative; proportional (quota share, surplus) and non-proportional (excess of loss).

Unit 3 — Income Tax (Questions 21–38)

Note on tax rates used: For tax calculations below I used the official tax rate references for Assessment Year 2025–26 (FY 2024–25) as published by the Indian tax authorities (health & education cess 4% on tax + surcharge). Where the new regime slab applies, I used the notified new-regime slab structure for FY 2024–25. [Income Tax India+1](#)

Worked questions first (full solutions).

21. From details of Mr. Nitin for FY 2024-25 — Evaluate taxable business income and tax payable (Old regime)

Given: Gross receipts ₹75,00,000; Expenses ₹44,00,000; Depreciation ₹7,00,000; Salary to staff ₹6,00,000; Rent ₹6,00,000.

Calculation of taxable business income:

- Total deductible expenses = ₹44,00,000 + ₹6,00,000 (salary) + ₹6,00,000 (rent) + ₹7,00,000 (depr) = ₹63,00,000.
- Taxable business income = Gross receipts – Total expenses = ₹75,00,000 – ₹63,00,000 = **₹12,00,000**.

Tax payable (Old regime, individual rates):

- Using standard slab: 0–2.5L nil; 2.5–5L @5%; 5–10L @20%; >10L @30%.
- Tax computation on ₹12,00,000:
 - 2.5–5.0L → ₹2,50,000 @5% = ₹12,500
 - 5.0–10.0L → ₹5,00,000 @20% = ₹1,00,000
 - 10.0–12.0L → ₹2,00,000 @30% = ₹60,000
 - **Total tax = ₹1,72,500.**
 - Add Health & Education Cess (4%) = ₹6,900.
 - **Total tax payable = ₹1,79,400.**

(No surcharge as income ≤ ₹50 lakh.) [Income Tax India](#)

22. Same data — tax payable (New regime)

- Taxable income (no additional deductions allowed) = **₹12,00,000**.
- New-regime slabs for FY 2024–25 (used here) typically give lower marginal rates but remove most exemptions/deductions — compute as: (Using the notified new-regime slab breakdown — calculation below.)
[Income Tax India](#)

Example new-regime computation (typical FY24-25 structure):

- 0–3L: nil
- 3–6L: ₹3,00,000 @5% = ₹15,000
- 6–9L: ₹3,00,000 @10% = ₹30,000
- 9–12L: ₹3,00,000 @15% = ₹45,000
- **Total tax (new regime) = ₹90,000.** Add cess 4% = ₹3,600 → **Total = ₹93,600.**

Conclusion: Under these numbers the **new regime (≈₹93,600)** is substantially better than the old regime (₹1,79,400) because of lower slab rates even though there are fewer deductions. [Income Tax India](#)

23. Explain: Heads of Income

Short note: Income is classified under five heads for tax: (1) Income from Salary, (2) Income from House Property, (3) Profits & Gains of Business or Profession, (4) Capital Gains (short & long term), and (5) Income from Other Sources. Each head has its own computation rules and allowable deductions.

24. Mrs. Ahuja (48) — Salary ₹7,85,000; Interest from RD ₹15,000; Deduction u/s 80C ₹1,20,000. Compute Total Income & Tax Payable (Old Regime).

- Gross income = ₹7,85,000 + ₹15,000 = ₹8,00,000.
- Deductions (80C) = ₹1,20,000.
- Taxable income = ₹8,00,000 – ₹1,20,000 = ₹6,80,000.
Tax (old slab):
 - 2.5–5.0L → ₹2,50,000 @5% = ₹12,500
 - 5.0–6.8L → ₹1,80,000 @20% = ₹36,000
→ Tax = ₹48,500. + cess 4% = ₹1,940 → **Total = ₹50,440.**

25. Same facts — New Regime

- No 80C under new regime (unless opted exceptions), so taxable income = ₹8,00,000.
- New-regime calculation (example slabs):
 - 3–6L → ₹3,00,000 @5% = ₹15,000
 - 6–8L → ₹2,00,000 @10% = ₹20,000
→ Tax = ₹35,000; cess 4% = ₹1,400 → **Total = ₹36,400.**

Conclusion: New regime ≈ ₹36,400 vs Old ≈ ₹50,440 → **new regime better** for these numbers. (Use official slab table for exact authority.) [Income Tax India](#)

26. Define: Assessee

Short note: An assessee is a person by whom any tax or any other sum of money is payable under the Income Tax Act. It includes individual, HUF, company, firm, AOP, BOI, local authority, or any other association/body.

27. Explain: Previous Year

Short note: Previous Year is the financial year (FY) in which income is earned. For AY 2025–26 the previous year is FY 2024–25 (1 Apr 2024–31

Mar 2025). Income of the previous year is assessed in the following Assessment Year.

28.Explain: Assessment Year

Short note: Assessment Year (AY) is the year following the previous year in which income of the previous year is assessed (e.g., income earned in FY 2024–25 is assessed in AY 2025–26).

29.Define: Income

Short note: Income means any profit, gain, wages, receipts, or any benefit, whether in money or kind, which is chargeable under the Income Tax Act and arises from any source.

30.Write a note on Surcharge with applicable rates

Short note: Surcharge is an additional tax on income-tax, applicable when total income crosses thresholds. Rates vary (e.g., 10% for incomes above ₹50 lakh/₹1 crore in some cases, higher slabs for very high incomes). Exact percentages and marginal relief rules change by year — check official tax tables. Health & Education Cess (4%) is charged on tax + surcharge. [Income Tax India+1](#)

31.Explain: Short Term Capital Gain

Short note: STCG arises when a capital asset is held for short period before transfer (period differs by asset: e.g., listed securities <12 months earlier — special rates). Tax treatment varies: some STCG taxed at special rates (e.g., 15% for securities under section 111A), others taxed at slab rates.

32.Explain: Long Term Capital Gain

Short note: LTCG arises when asset held beyond specified holding period (e.g., >24/36 months for some assets or >12 months for listed equity depending on asset type). LTCG on listed equity over ₹1 lakh is taxed at 10% without indexation (subject to conditions) — other assets have different rates.

33.Explain: Income from Salary

Short note: Salary income includes basic pay, allowances (some fully/partially exempt), perquisites, bonuses. Taxable salary is gross salary minus exemptions/allowances and standard deduction (if applicable). Employer's PF contribution and perquisites may have specific valuation rules.

**34. Mr. Chintan (56) — Annual salary ₹15,00,000, deductions ₹2,50,000.
Calculate income tax in both old and new slabs and suggest best slab.**

- **Old regime:** Taxable income = ₹15,00,000 – ₹2,50,000 = **₹12,50,000.**

Tax (old slabs):

- 2.5–5.0L = ₹2.5L @ 5% = ₹12,500
- 5.0–10.0L = ₹5.0L @ 20% = ₹1,00,000
- 10.0–12.5L = ₹2.5L @ 30% = ₹75,000

→ Tax = ₹1,87,500. Add cess 4% (₹7,500) → **Total = ₹1,95,000.**

- **New regime:** No deductions, taxable = **₹15,00,000.**

New-regime compute (example slabs used):

- 3–6L: ₹3L @ 5% = ₹15,000
- 6–9L: ₹3L @ 10% = ₹30,000
- 9–12L: ₹3L @ 15% = ₹45,000
- 12–15L: ₹3L @ 20% = ₹60,000

→ Tax = ₹1,50,000. Add cess 4% = ₹6,000 → **Total = ₹1,56,000.**

Conclusion: New regime (≈₹1,56,000) is better than old (≈₹1,95,000) for these numbers — **choose new regime.**

36. Mr. Shyam — Indian citizen employed in Canada for 12 years, returned and stayed for 185 days in previous year. Determine residential status for AY 2025–26.

Short note & answer: Under Indian tax law, residential status depends on days present in India during previous year: ordinarily, ≥182 days (now 120 or 182 depending on conditions) can make you resident. As he stayed **185 days**, he satisfies basic residence test → **Resident** for AY 2025–26 (subject to other ties and exact rules like “resident but not ordinarily resident” if conditions for NRI history apply).

37. When it is mandatory to file Income Tax Return?

Short note: Mandatory when gross total income exceeds basic exemption limit, when refund claim, if taxable under special heads, if subject to tax audit, if foreign assets, or specified transactions occur (e.g., deposit thresholds). Exact thresholds and conditions vary — check current IT provisions.

38. Write down new slab rates applicable to individuals for FY 2024–25.

Short note: New regime slabs for FY 2024–25 were notified by the government — main points: lower marginal rates across slabs (starting

with nil up to a lower threshold, then 5%, 10%, 15%, 20%, 30%) and removal of many deductions/exemptions. Always verify with current official tax-rate table. [Income Tax India](#)

Unit 4 — GST (Questions 40–56)

40.Explain: Need of GST

Short note: GST replaced cascading indirect taxes (central & state) with a unified tax on supply of goods & services to simplify taxation, reduce tax-on-tax, provide uniform tax base, enlarge compliance through IT systems, and enable seamless input tax credit across supply chain.

41.Explain: Input Tax Credit

Short note: ITC allows registered persons to claim credit of GST paid on inputs/services used for making taxable supplies. Conditions: tax invoice, receipt of goods/services, tax paid by supplier (or payable in case of reverse charge), return filed. ITC reduces cascading and final tax liability.

42.Explain: Time of Supply

Short note: Time of supply determines when tax becomes payable — typically earlier of invoice date or receipt of payment (and for goods in movement, date of removal). Special rules apply for continuous supply and reverse charge.

43.Explain: Place of Supply

Short note: Place of supply rules determine whether supply is intra-state (CGST+SGST) or inter-state (IGST). For goods, usually where goods are delivered; for services, rules depend on supplier/recipient locations and nature of service.

44.A Ltd: Purchase raw material ₹5,00,000 + GST @10%. Sold finished goods ₹6,50,000 + GST @10%. Find: Output Tax, Input Tax Credit and Net Tax Payable.

- Input tax = 10% of ₹5,00,000 = **₹50,000**.
- Output tax = 10% of ₹6,50,000 = **₹65,000**.
- Net tax payable after ITC = ₹65,000 – ₹50,000 = **₹15,000**.

45.A Ltd: Purchase ₹8,00,000 + GST@20%. Sold ₹10,00,000 + GST@20%.

- Input tax = 20% of ₹8,00,000 = **₹1,60,000**.
- Output tax = 20% of ₹10,00,000 = **₹2,00,000**.
- Net payable = ₹2,00,000 – ₹1,60,000 = **₹40,000**.

46. How does the GST system work?

Short note: GST is collected at each stage of supply with credit for tax paid on purchases (ITC). Supplier charges output GST, remits net GST after adjusting ITC. Distinguishes intra-state (CGST+SGST) and inter-state (IGST).

47. Saral Ltd. example — GST at each stage (18%) and net tax payable after ITC.

Given flow:

- Raw material supplier → Manufacturer (raw material value ₹2,00,000 @18% GST)
- Manufacturer adds value ₹1,00,000 → sells to Wholesaler at ₹3,00,000 @18%
- Wholesaler adds ₹1,00,000 → sells to Retailer at ₹4,00,000 @18%
- Retailer sells to Consumer at ₹4,80,000 @18%

Compute stage-wise GST and net payable (all values are GST on sale value):

1. Raw material purchase (manufacturer pays input GST)

- GST on ₹2,00,000 = ₹36,000 (input tax for manufacturer).

2. Manufacturer sells to wholesaler

- Sale value = ₹2,00,000 + ₹1,00,000 = ₹3,00,000. Output GST = 18% of ₹3,00,000 = ₹54,000.
- Manufacturer ITC = ₹36,000 → Net GST payable by manufacturer = ₹54,000 – ₹36,000 = **₹18,000**.

3. Wholesaler sells to retailer

- Sale value = ₹3,00,000 + ₹1,00,000 = ₹4,00,000. Output GST = 18% of ₹4,00,000 = ₹72,000.
- Wholesaler ITC (from manufacturer's invoice) = ₹54,000 → Net GST payable by wholesaler = ₹72,000 – ₹54,000 = **₹18,000**.

4. Retailer sells to consumer

- Sale value = ₹4,80,000. Output GST = 18% of ₹4,80,000 = ₹86,400.
- Retailer ITC = ₹72,000 → Net GST payable by retailer = ₹86,400 – ₹72,000 = **₹14,400.**

Total net GST collected by Government = ₹18,000 + ₹18,000 + ₹14,400 = ₹50,400.

(Stage-wise ITC ensures only tax on value addition is borne at each stage.)

48. Savitri Ltd example — GST @9%

Given: Raw material ₹8,00,000 + 9% GST; manufacturer adds ₹4,00,000; wholesaler adds ₹4,00,000; final sale to consumer quoted as ₹9,60,000.

Important: The numbers are inconsistent (if RM = ₹8,00,000 and two stages of ₹4,00,000 additions, final pre-tax value should be ₹16,00,000 not ₹9,60,000). Because of this mismatch, I'll compute GST stage-wise based on the stage values (RM → manufacturer → wholesaler → retailer) and then note the inconsistency.

Stagewise (using given stage values):

- RM GST = 9% of ₹8,00,000 = **₹72,000** (input tax).
- Manufacturer sells at ₹8,00,000 + ₹4,00,000 = ₹12,00,000 → output GST = 9% of ₹12,00,000 = **₹1,08,000** → Net payable = 1,08,000 – 72,000 = **₹36,000.**
- Wholesaler sells at ₹12,00,000 + ₹4,00,000 = ₹16,00,000 → output GST = 9% of ₹16,00,000 = **₹1,44,000** → Net payable = 1,44,000 – 1,08,000 = **₹36,000.**
- Retailer's sale to consumer: if it truly were ₹16,00,000 then GST = 1,44,000 and net payable = 1,44,000 – 1,44,000 = 0. But the problem states final sale is **₹9,60,000**, which contradicts stage values — if final sale is ₹9,60,000, recomputation of the wholesaler/retailer values versus the earlier stages is required.

Conclusion & note: Because the question contains inconsistent numbers (intermediate cumulative value = ₹16,00,000 but final consumer price is given as ₹9,60,000), I computed GST on the consistent stage values and showed net payable at each stage. If you want calculations **adjusted to a final consumer**

price of ₹9,60,000, tell me which stage values should be revised and I'll recompute accordingly.

49.Explain: GSTR-1

Short note: GSTR-1 is the monthly/quarterly return (by small taxpayers) filed by suppliers containing details of outward supplies/sales (invoices). It supplies invoice-level data for recipients to reconcile ITC.

50.Explain: GSTR-3B

Short note: GSTR-3B is a monthly self-declaration summary return for payment of GST (tax liabilities, ITC claimed). It's simpler but mandatory for all regular taxpayers to deposit tax and report supplies.

51.Explain: Supply under GST

Short note: 'Supply' is broadly defined as sale, transfer, barter, exchange, license, rental, lease, or disposal of goods/services for consideration — includes both taxable & exempt supplies. Place/time/value/supply classification rules apply.

52.Explain: Composite Supply

Short note: Composite supply consists of two or more taxable supplies naturally bundled where one supply is principal. Tax rate of the principal supply applies to the whole bundle.

53.Explain: Mixed Supply

Short note: Mixed supply = two or more supplies made together that are not naturally bundled; the highest GST rate among the supplies applies to the entire mixed supply.

54.Compare: GSTR-2B and GSTR-3B

Short note: GSTR-2B is an auto-drafted ITC statement (monthly) generated for recipients — static and used to reconcile eligible ITC. GSTR-3B is the self-declared monthly summary return taxpayers file to discharge tax liability. 2B helps fill 3B but 3B is the payment return.

55.Mr. Sohan (distributor) & Mr. Mohan (retailer) — invoice timing Qs

Given: movement of goods on 2 March 2025; delivery on 1 March 2025.
Answer: Under GST, invoice must generally be issued before or at the time of removal/delivery (time of supply rules). If goods moved on 2 March 2025, invoice should reflect correct date — raising invoice before delivery (i.e., before goods are handed over) can be done in some cases

(e.g., advance invoices), but the invoice date should align with time of supply rules. Practically: invoice for sale should be dated on or before the time of supply (check specific rule applying to who bears responsibility). (Short answer: invoice date must conform to time-of-supply; you cannot arbitrarily back-date an invoice.)

56. What are limits for GST registration?

Short note: Registration threshold: generally aggregate turnover above ₹20 lakh (₹10 lakh for special category states) for goods/services (thresholds can differ by supply type/Inter-state/online etc.). For certain businesses (inter-state supply, e-commerce, reverse charge), registration is mandatory irrespective of threshold. (Confirm current thresholds on GST portal for exact numbers.)

Unit 5 — Financial Management / Time Value of Money (Questions 57–65)

57. Explain concept of time value of money.

Short note: Money today is worth more than the same nominal amount in the future due to its earning potential (interest), inflation, and risk. Present Value (PV) discounts future cash flows; Future Value (FV) compounds current money.

58. Reasons for preference of money.

Short note: Prefer money now because (1) opportunity to invest and earn returns, (2) consumption preference (present consumption utility), (3) inflation reduces future purchasing power, (4) risk & uncertainty over future receipts.

59. Practical sum of Present value (Single Amount and Annuity)

- **PV of single future amount:** $PV = \frac{FV}{(1+r)^n}$

$$PV = \frac{FV}{(1+r)^n}$$

Example: PV of ₹1,00,000 to be received in 3 years at 8% p.a. →

$$PV = \frac{100000}{(1.08)^3} = \frac{100000}{1.259712} \approx ₹79,383$$

$$PV = \frac{100000}{1.259712} \approx ₹79,383$$

- **PV of an ordinary annuity (equal payments at period end):**

$$PV = P \times \frac{1 - (1+r)^{-n}}{r}$$

Example: PV of ₹10,000/year for 5 years at 6%:

$$PV = 10000 \times \frac{1 - (1.06)^{-5}}{0.06} \approx 10000 \times 4.21236 = ₹42,123.6$$

$$PV = 10000 \times \frac{1 - (1.06)^{-5}}{0.06} \approx 10000 \times 4.21236 = ₹42,123.6$$

60. Practical sum of Future value (Single Amount and Annuity)

- FV of single amount:** $FV = PV \times (1+r)^n$
 Example: ₹50,000 invested 4 years at 7% → $FV = 50,000 \times (1.07)^4 = 50,000 \times 1.3108 \approx ₹65,540$.
- FV of ordinary annuity (payments at period end):**
 $FV = P \times \frac{(1+r)^n - 1}{r}$
 Example: ₹5,000/year for 10 years at 8% → $FV = 5,000 \times \frac{(1.08)^{10} - 1}{0.08} \approx 5,000 \times 14.4866 = ₹72,433$.

61. Practical sum of Sinking Fund, Intra Year Compounding and Discounting

- Sinking fund (to accumulate a target future amount):** payment AAA required when target FV is SSS:
 $A = S \times \frac{r}{(1+r)^n - 1}$
 Example: Want ₹1,00,000 in 5 years at 6%:
 $A = 100000 \times 0.06 / ((1.06)^5 - 1) \approx 100000 \times 0.06 / 0.338225 \approx ₹17,745$
 $A = 100000 \times 0.06 / ((1.06)^5 - 1) \approx 100000 \times 0.06 / 0.338225 \approx ₹17,745$ per year.
- Intra-year compounding:** If compounding m times per year, effective rate $(1+r/m)^m - 1$. Use $FV = PV \times (1+r/m)^{mn}$. For discounting use the reciprocal.
- Discounting (finding PV):** For cash flows within the year or irregular flows, discount at appropriate fractional periods (e.g., for t months use $(1+r)^{t/12}$).

62. Practical sum of Annuity Due

- Annuity due (payments at period start):** PV and FV are adjusted by factor $(1+r)$ relative to ordinary annuity.
- $PV_{\text{due}} = PV_{\text{ordinary}} \times (1+r)$
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- $FV_{due} = FV_{ordinary} \times (1+r)$
 $FV_{due} = FV_{ordinary} \times (1+r)$
 $= FV_{ordinary} \times (1+r)$
 Example: ₹10,000 at start of year for 5 years at 5% → $PV_{due} = (10,000 \times (1 - (1.05)^{-5}) / 0.05) \times 1.05 \approx 10k \times 4.32948 \approx ₹43,294.8$.

63. Practical sum of Loan Amortization

- Loan amortization uses an EMI to repay principal+interest. EMI formula:
 $EMI = P \times r(1+r)^n / (1+r)^n - 1$
 $EMI = P \times \frac{r(1+r)^n}{(1+r)^n - 1}$
 $EMI = P \times (1+r)^n - 1 / (1+r)^n$ (r = periodic rate).
- Produce amortization schedule showing interest portion (opening balance × r) and principal portion (EMI – interest), and reduce balance.
 Example: Loan ₹5,00,000, 10% p.a., 5 years (annual comp): r=0.10, n=5
 → $EMI \approx 5,00,000 \times (0.1 \times 1.61051) / (1.61051 - 1) \approx 5,00,000 \times 0.161051 / 0.61051 \approx ₹131,501$ per year. (Interest/principal breakdown in schedule.)

64. Explain concept of Capital Recovery and Loan Amortization

Short note: Capital recovery is converting series of returns into equivalent annual amounts to recover the invested capital plus return.
 Loan amortization is scheduled repayments (EMI) that allocate payments between interest and principal so that loan becomes zero at maturity.

65. Discuss the Rule of Doubling

Short note: Rule of 72 (approx) — estimate years to double investment:
 Years $\approx 72 / (\text{annual interest rate in \%})$. Example: at 8% p.a., doubling $\approx 72/8 = 9$ years. (Exact doubling uses $\ln(2)/\ln(1+r)$.)